CSW332_Assignment4_PhamTranGiaHung

May 8, 2026

[1]:
```python
#question 1
def greet():
    print("Welcome to Python Programming")
greet()
```

```
Welcome to Python Programming
```

[29]:
```python
#question 2
def show_name(name):
    print(f"Hello, {name}! Greeting.")

show_name("Zhuang Fangyi")
```

```
Hello, Zhuang Fangyi! Greeting.
```

[31]:
```python
#question 3
def square(n):
    return n*n

print(square(5))
```

```
25
```

[34]:
```python
#question 4
def cube(n):
    return n*n*n

print(cube(5))
```

```
125
```

[35]:
```python
#question 5
def add(a, b):
    return a +b

print(add(4,7))
```

```
11
```

[73]:
```python
#question 6
def largest(a, b):
    return a if a > b else b

print(largest(10, 20))
```

20

[40]:
```python
#question 7
def even_odd(n):
    if n % 2 == 0:
        return "Even"
    else:
        return "Odd"

print(even_odd(8))
```

Even

[41]:
```python
#question 8
def factorial(n):
    result = 1
    for i in range(1, n + 1):
        result *= i
    return result

print(factorial(5))
```

120

[64]:
```python
#question 9
def average(a, b, c):
    return (a + b + c) / 3

print(average(10,20,30))
```

20.0

[66]:
```python
#question 10
def count_digits(n):
    return len(str(abs(n)))

print(count_digits(122333666666)) #12
```

12

[68]:
```python
#question 11
def list_sum(lst):
    return sum(lst)
```

```
print(list_sum([1,3,5,7, 9]))
```

25

[45]:
```
#question 12
def largest_number(lst):
    return max(lst)

print(largest_number([10, 25, 3, 47, 18]))
```

47

[46]:
```
#question 13
def smallest_number(lst):
    return min(lst)

print(smallest_number([10, 25, 3, 47, 18]))
```

3

[79]:
```
#question 14
def count_even(lst):
    count = 0
    for x in lst:
        if x % 2 == 0:
            count += 1
    return count

print(count_even([1, 2, 5, 6, 8, 9 ,10])) #4
```

4

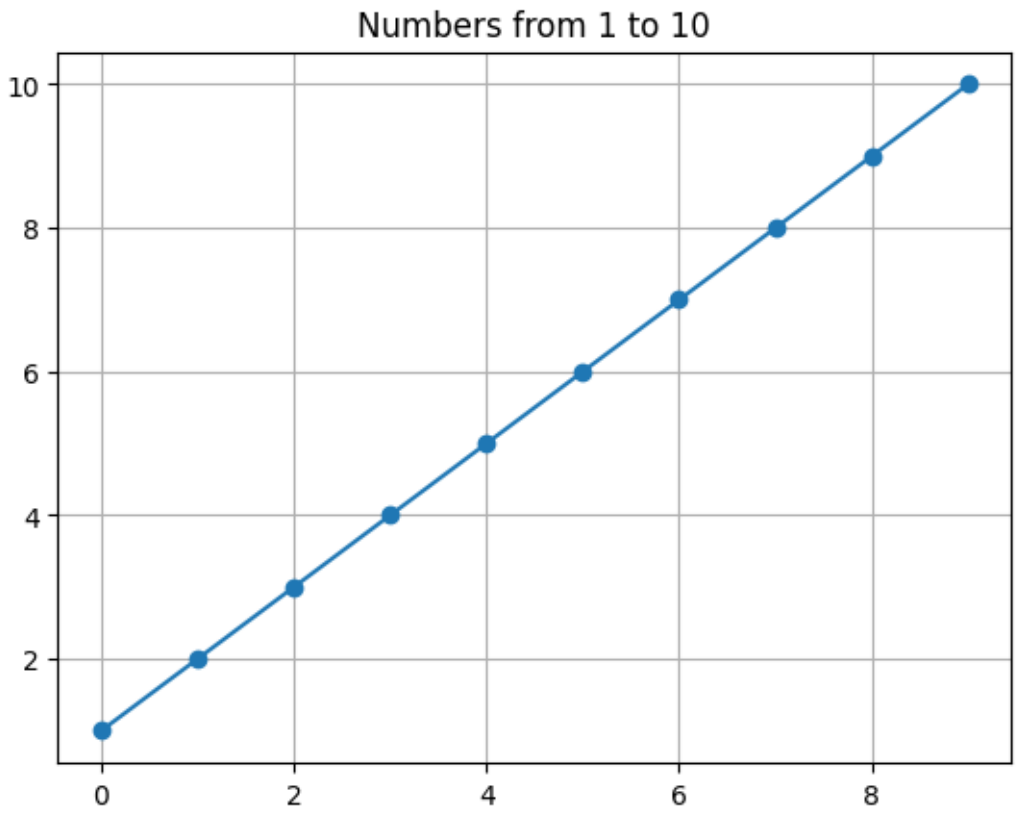
[80]:
```
#question 15
def reverse_list(lst):
    return lst[::-1]

print(reverse_list([1, 2, 3, 4, 5, 6]))
```

[6, 5, 4, 3, 2, 1]

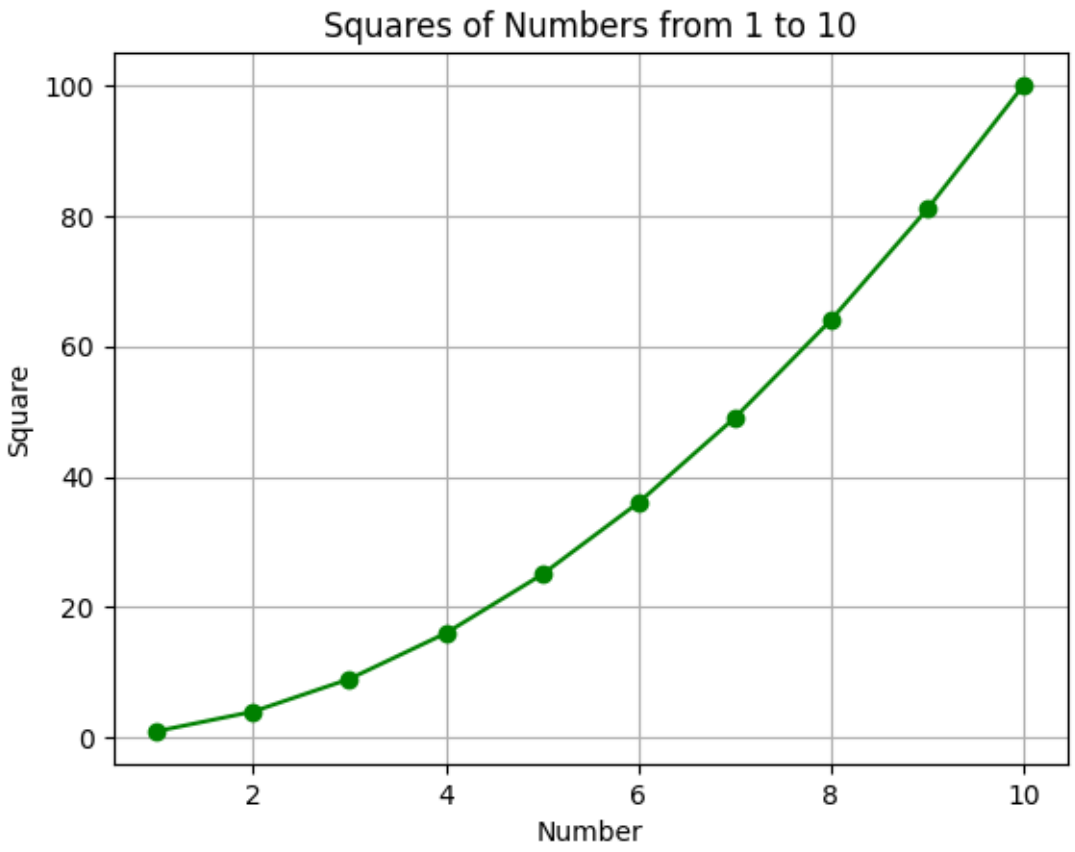
[86]:
```
#question 16
import matplotlib.pyplot as plt

numbers = list(range(1, 11))
plt.plot(numbers, marker='o')
plt.title("Numbers from 1 to 10")
plt.grid(True)
plt.show()
```



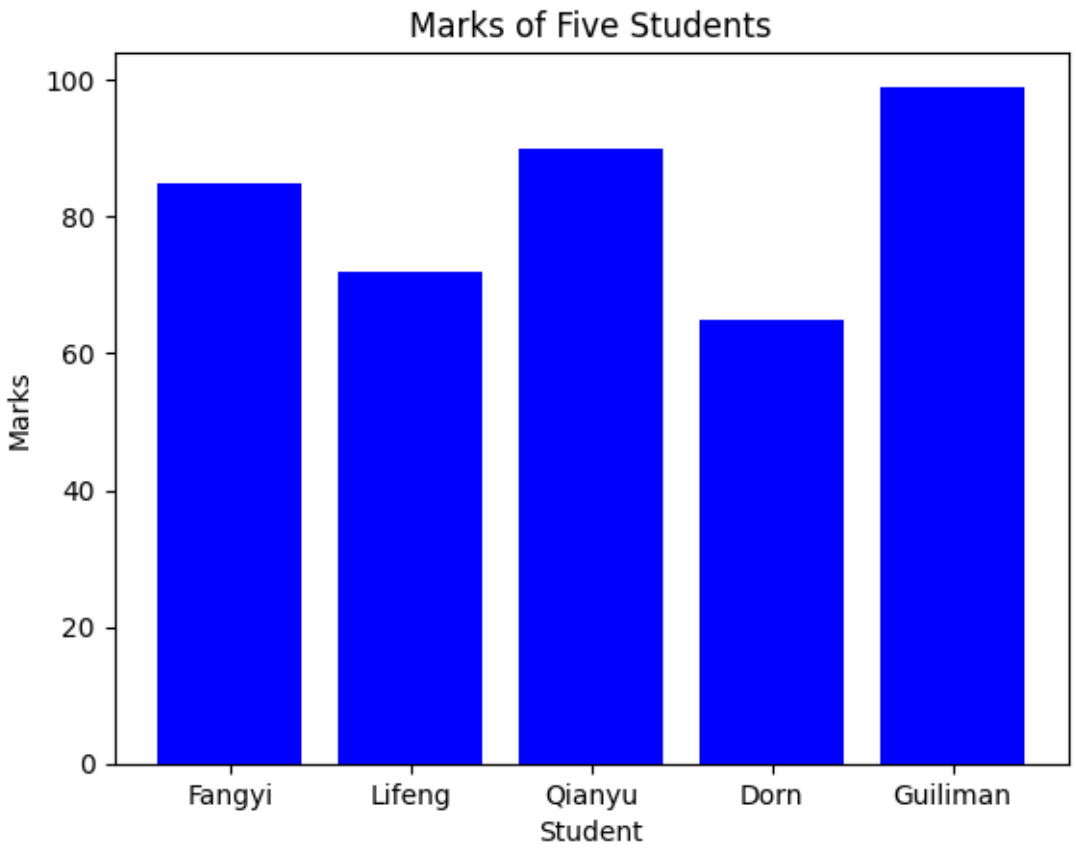
[50]:
```
#question 17
import matplotlib.pyplot as plt

numbers = list(range(1, 11))
squares = [n ** 2 for n in numbers]
plt.plot(numbers, squares, marker='o', color='green')
plt.title("Squares of Numbers from 1 to 10")
plt.xlabel("Number")
plt.ylabel("Square")
plt.grid(True)
plt.show()
```



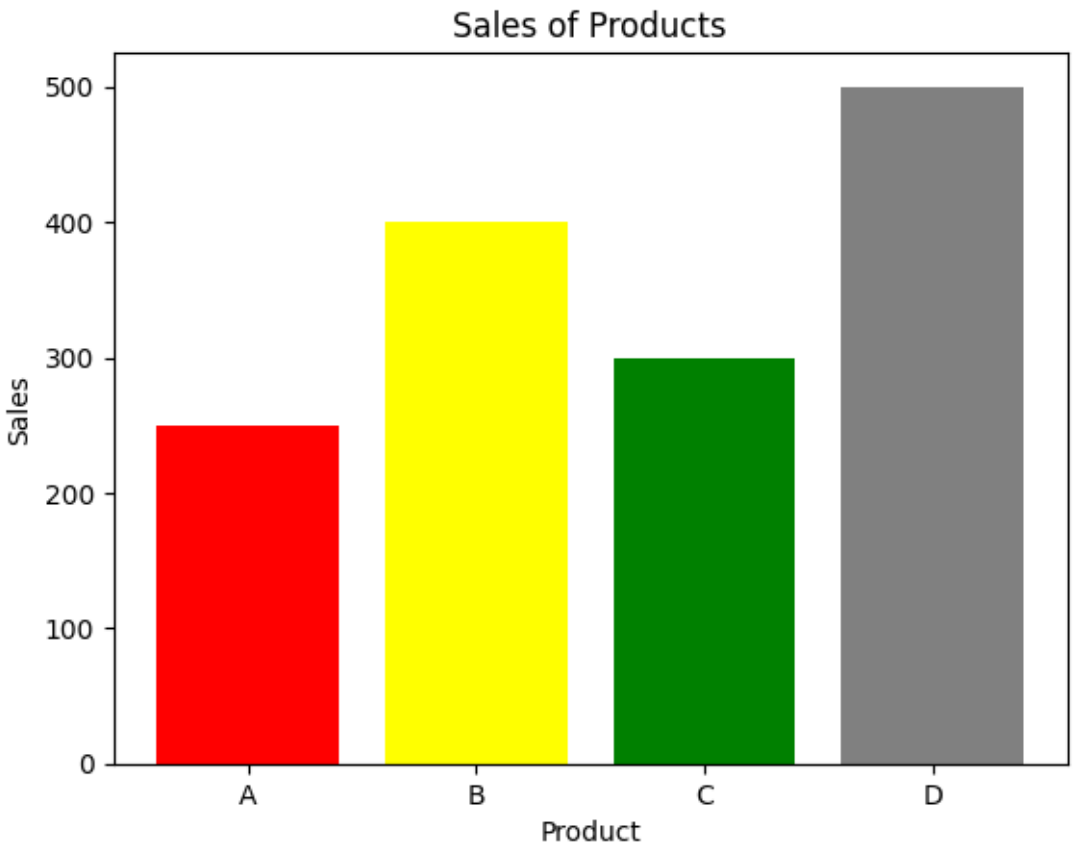
[1]:
```
#question 18
import matplotlib.pyplot as plt

students = ['Fangyi', 'Lifeng', 'Qianyu', 'Dorn', 'Guiliman']
marks = [85, 72, 90, 65, 99]
plt.bar(students, marks, color='blue')
plt.title("Marks of Five Students")
plt.xlabel("Student")
plt.ylabel("Marks")
plt.show()
```



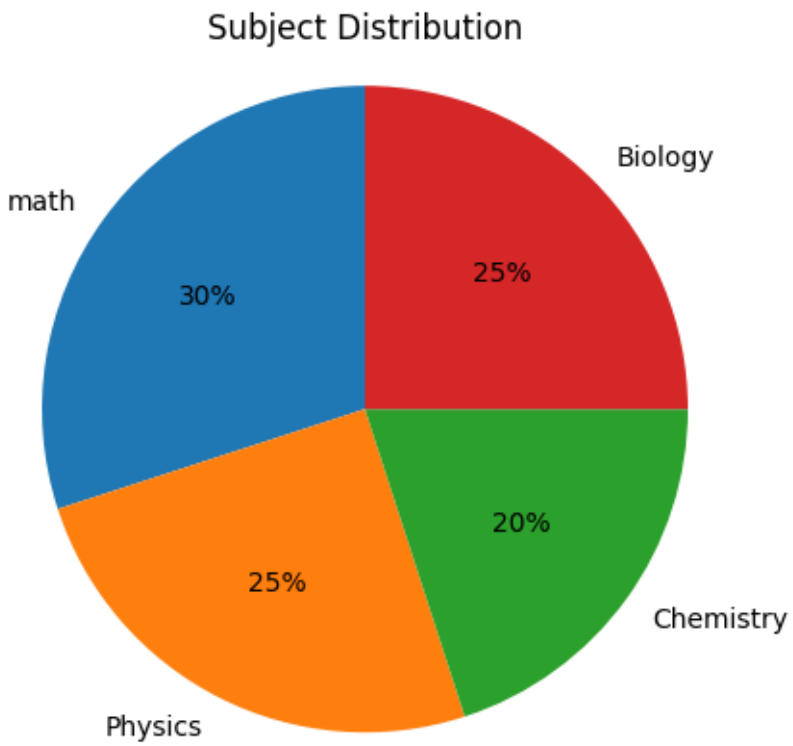
[2]:
```
#question 19
import matplotlib.pyplot as plt

products = ['A', 'B', 'C', 'D']
sales = [250, 400, 300, 500]
plt.bar(products, sales, color=['red', 'yellow', 'green', 'gray'])
plt.title("Sales of Products")
plt.xlabel("Product")
plt.ylabel("Sales")
plt.show()
```



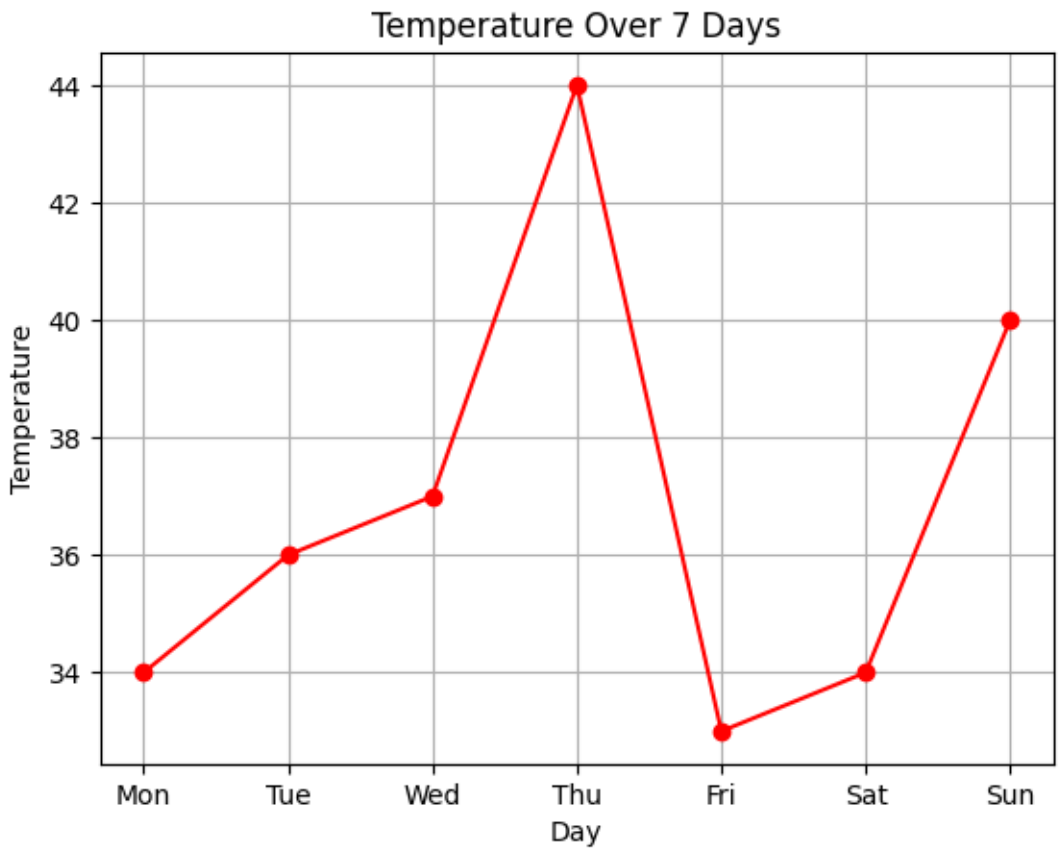
[9]:
```
#question 20
import matplotlib.pyplot as plt

subjects = ['math', 'Physics', 'Chemistry', 'Biology']
percentages = [30, 25, 20, 25]
plt.pie(percentages, labels=subjects, autopct='%1.0f%%',startangle=90)
plt.title("Subject Distribution")
plt.axis('equal')
plt.show()
```



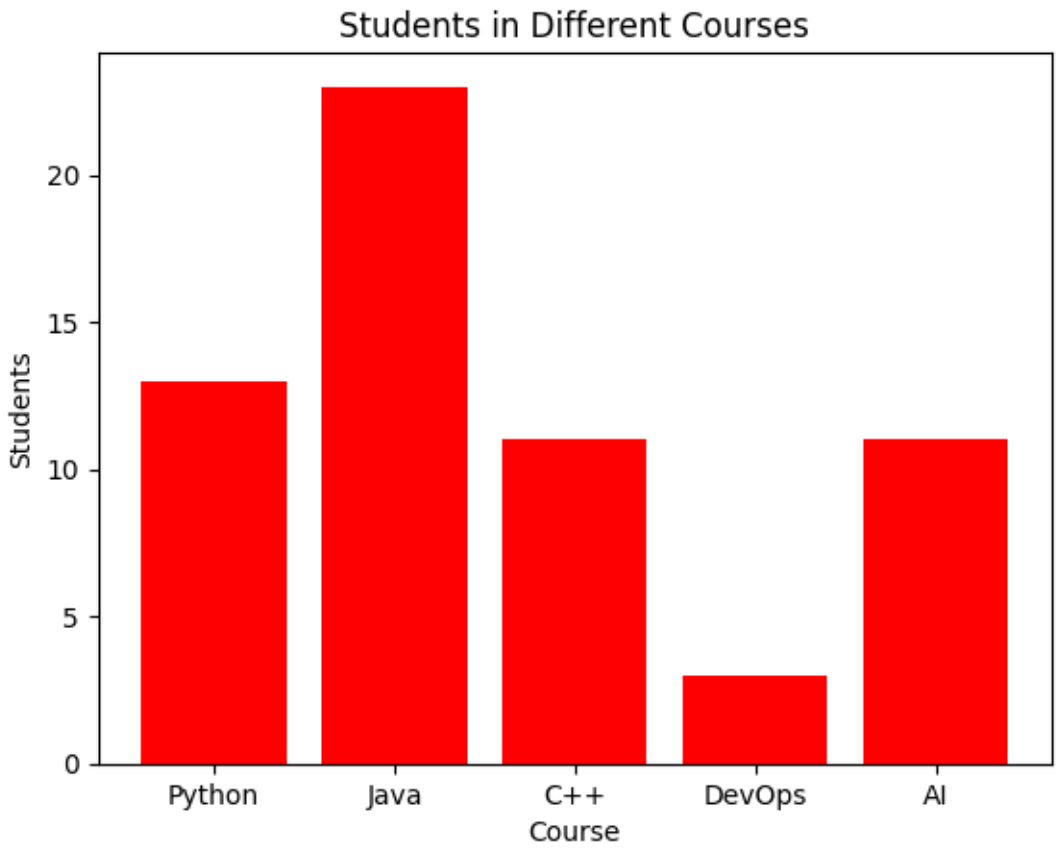
[8]:
```
#question 21
import matplotlib.pyplot as plt

days = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']
temperatures = [34, 36, 37, 44, 33, 34, 40]
plt.plot(days, temperatures, marker='o', color='red')
plt.title("Temperature Over 7 Days")
plt.xlabel("Day")
plt.ylabel("Temperature")
plt.grid(True)
plt.show()
```



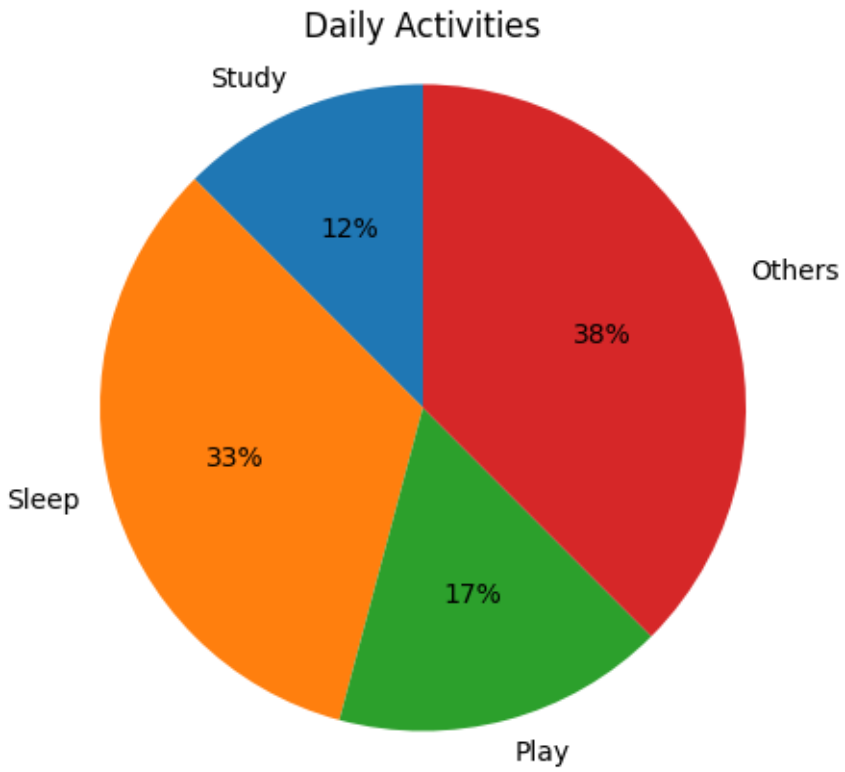
[11]:
```
#question 22
import matplotlib.pyplot as plt

courses = ['Python', 'Java', 'C++', 'DevOps', 'AI']
num_students = [13, 23, 11, 3, 11]
plt.bar(courses, num_students, color='red')
plt.title("Students in Different Courses")
plt.xlabel("Course")
plt.ylabel("Students")
plt.show()
```



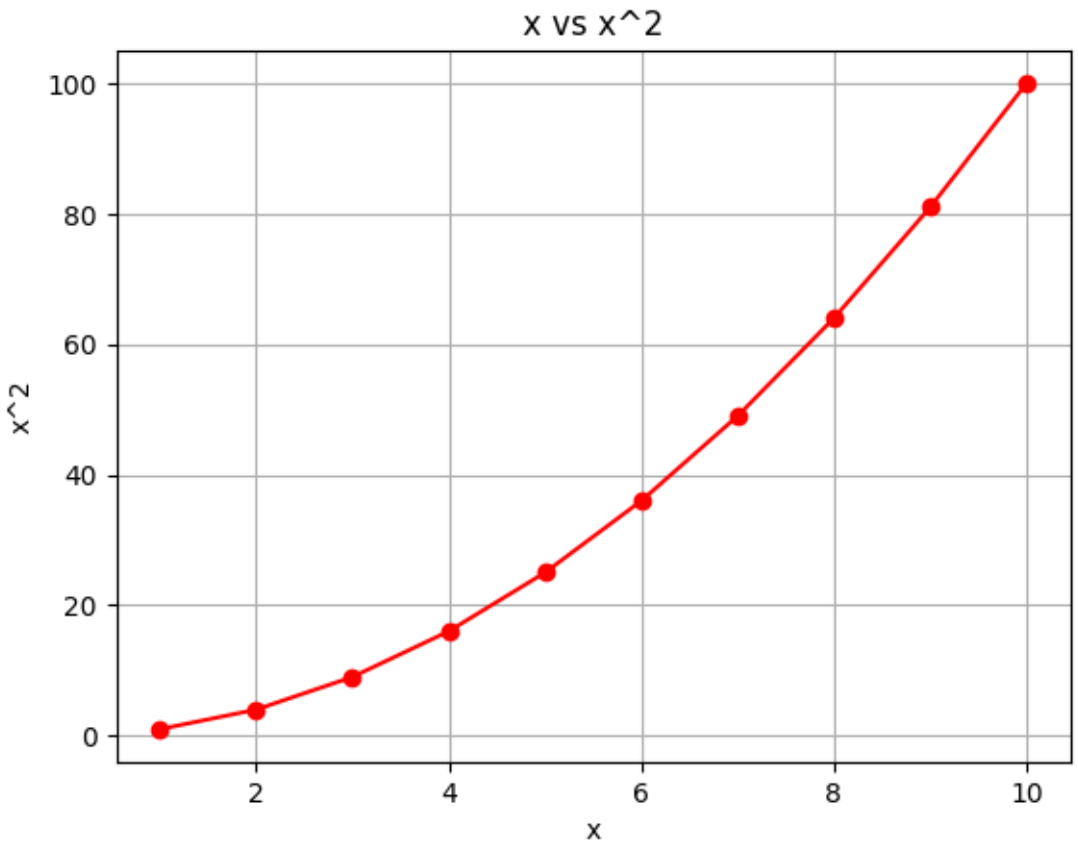
[13]:
```
#question 23
import matplotlib.pyplot as plt

activities = ['Study', 'Sleep', 'Play', 'Others']
hours = [3, 8, 4, 9]
plt.pie(hours, labels=activities, autopct='%1.0f%%', startangle=90)
plt.title("Daily Activities")
plt.axis('equal')
plt.show()
```



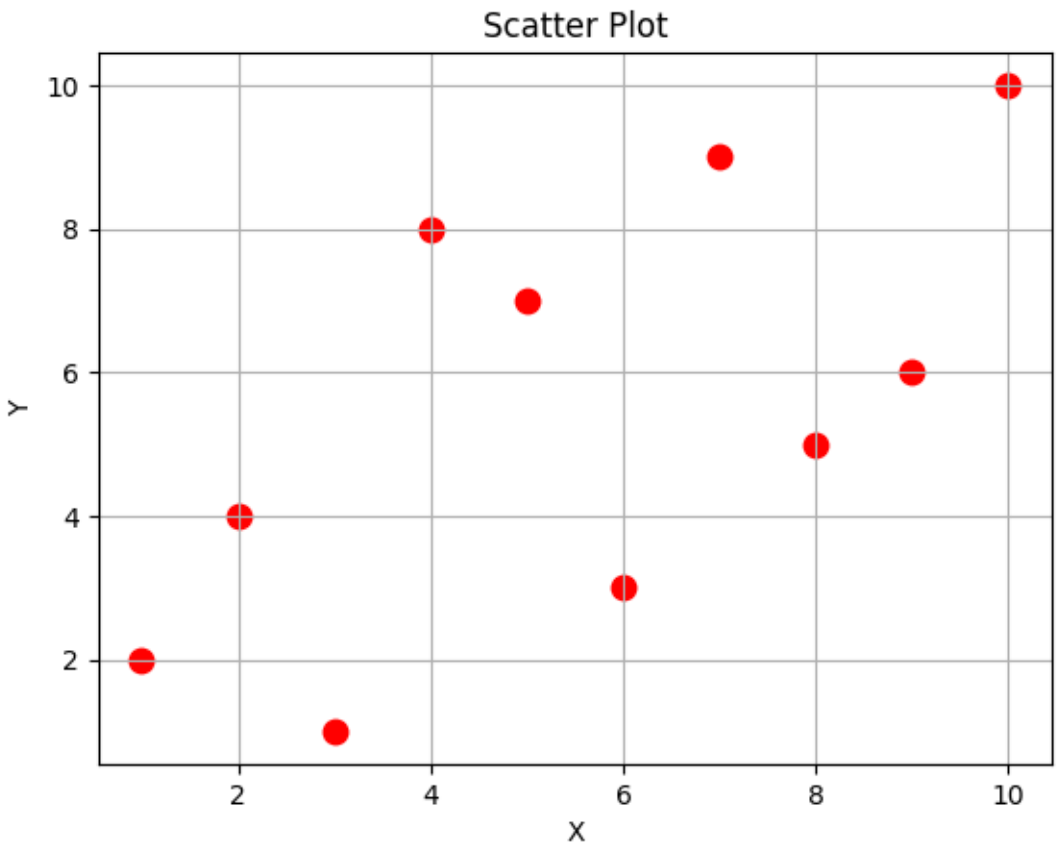
[16]:
```
#question 24
import matplotlib.pyplot as plt

x = list(range(1, 11))
x_squared = [i ** 2 for i in x]
plt.plot(x, x_squared, marker='o', color='red')
plt.title("x vs x^2")
plt.xlabel("x")
plt.ylabel("x^2")
plt.grid(True)
plt.show()
```



[17]:
```
#question 25
import matplotlib.pyplot as plt

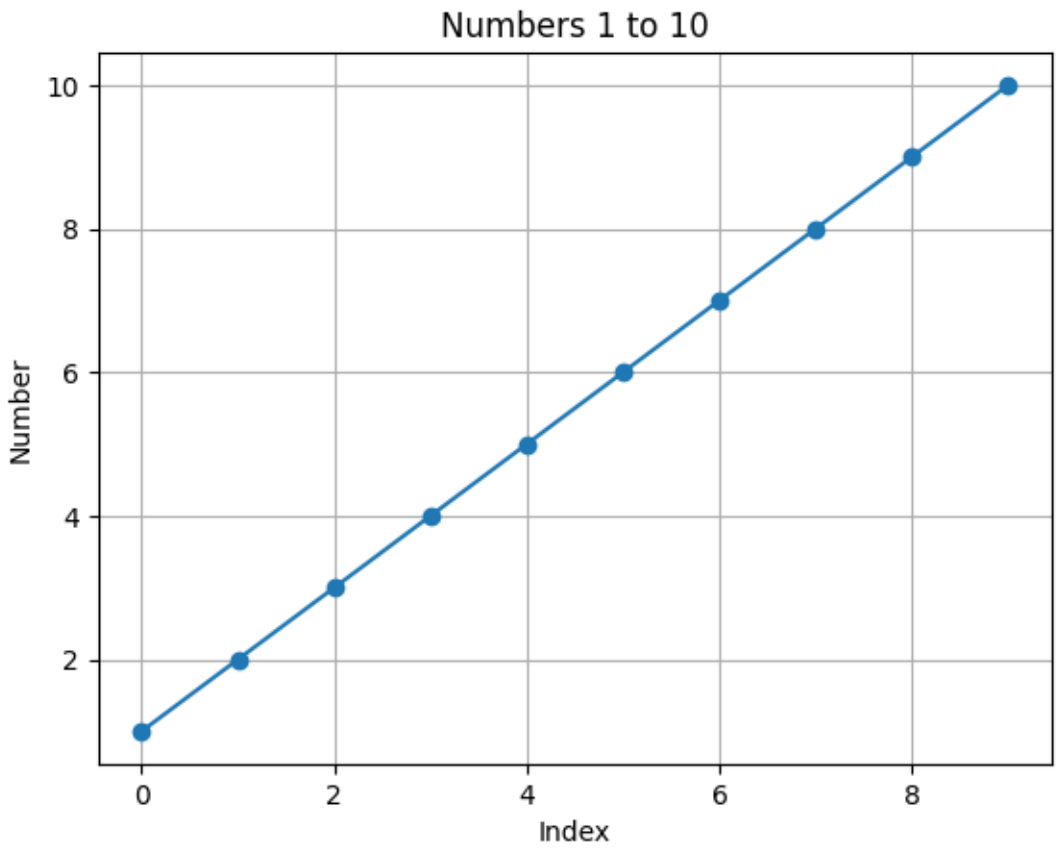
x = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
y = [2, 4, 1, 8, 7, 3, 9, 5, 6, 10]
plt.scatter(x, y, color='red', s=80)
plt.title("Scatter Plot")
plt.xlabel("X")
plt.ylabel("Y")
plt.grid(True)
plt.show()
```



[18]:
```python
#question 26
import matplotlib.pyplot as plt

def generate_and_plot():
    numbers = list(range(1, 11))
    plt.plot(numbers, marker='o')
    plt.title("Numbers 1 to 10")
    plt.xlabel("Index")
    plt.ylabel("Number")
    plt.grid(True)
    plt.show()

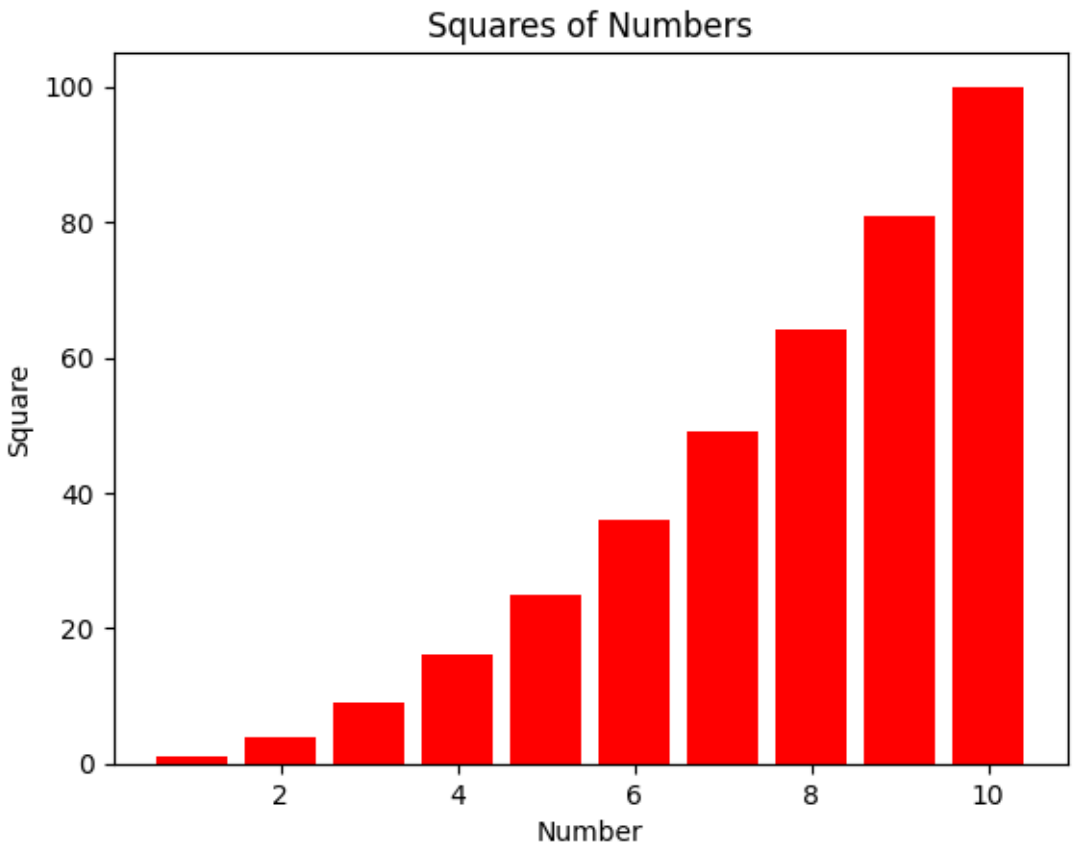
generate_and_plot()
```



[19]:
```
#question 27
import matplotlib.pyplot as plt

def squares_bar_chart(numbers):
    squares = [n ** 2 for n in numbers]
    plt.bar(numbers, squares, color='red')
    plt.title("Squares of Numbers")
    plt.xlabel("Number")
    plt.ylabel("Square")
    plt.show()

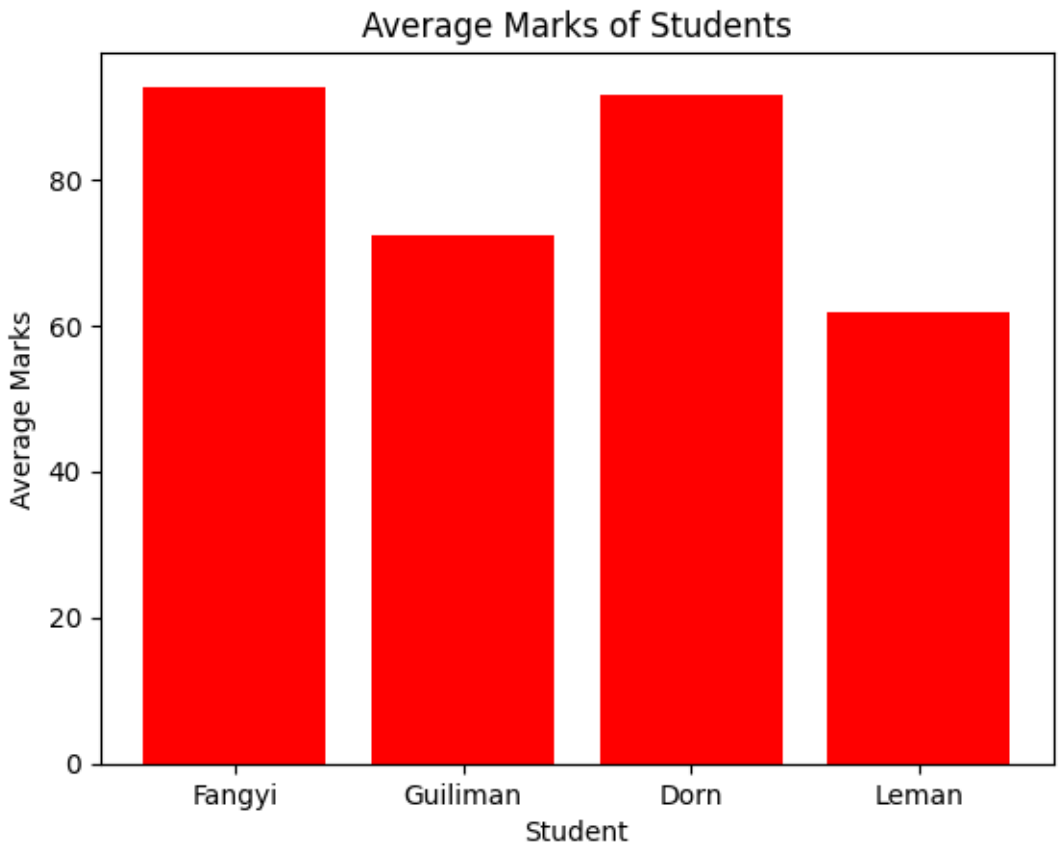
squares_bar_chart([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])
```



[21]:
```python
#question 28
import matplotlib.pyplot as plt

def average_marks_bar(student_marks):
    names = list(student_marks.keys())
    averages = [sum(marks) / len(marks) for marks in student_marks.values()]
    plt.bar(names, averages, color='red')
    plt.title("Average Marks of Students")
    plt.xlabel("Student")
    plt.ylabel("Average Marks")
    plt.show()

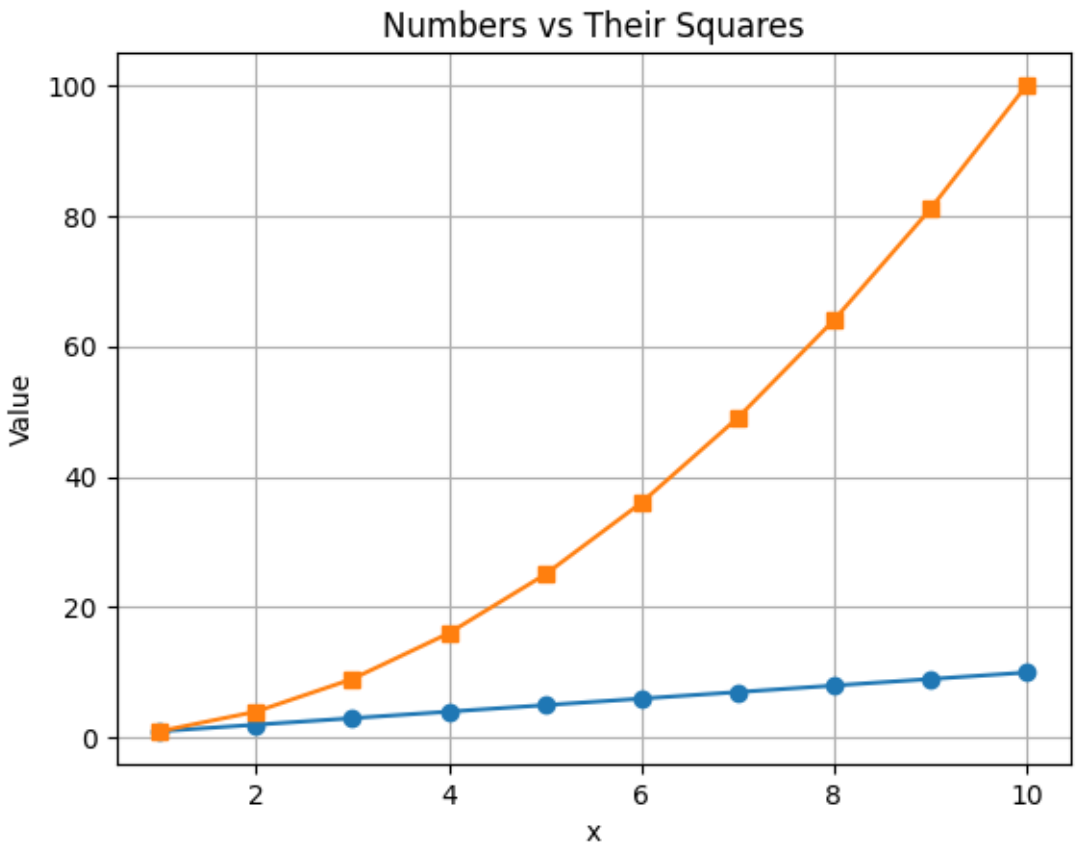
marks_data = {
    'Fangyi': [99, 90, 89],
    'Guiliman': [72, 65, 80],
    'Dorn': [88, 92, 95],
    'Leman': [60, 70, 55]
}
average_marks_bar(marks_data)
```



[24]:
```python
#question 29
import matplotlib.pyplot as plt

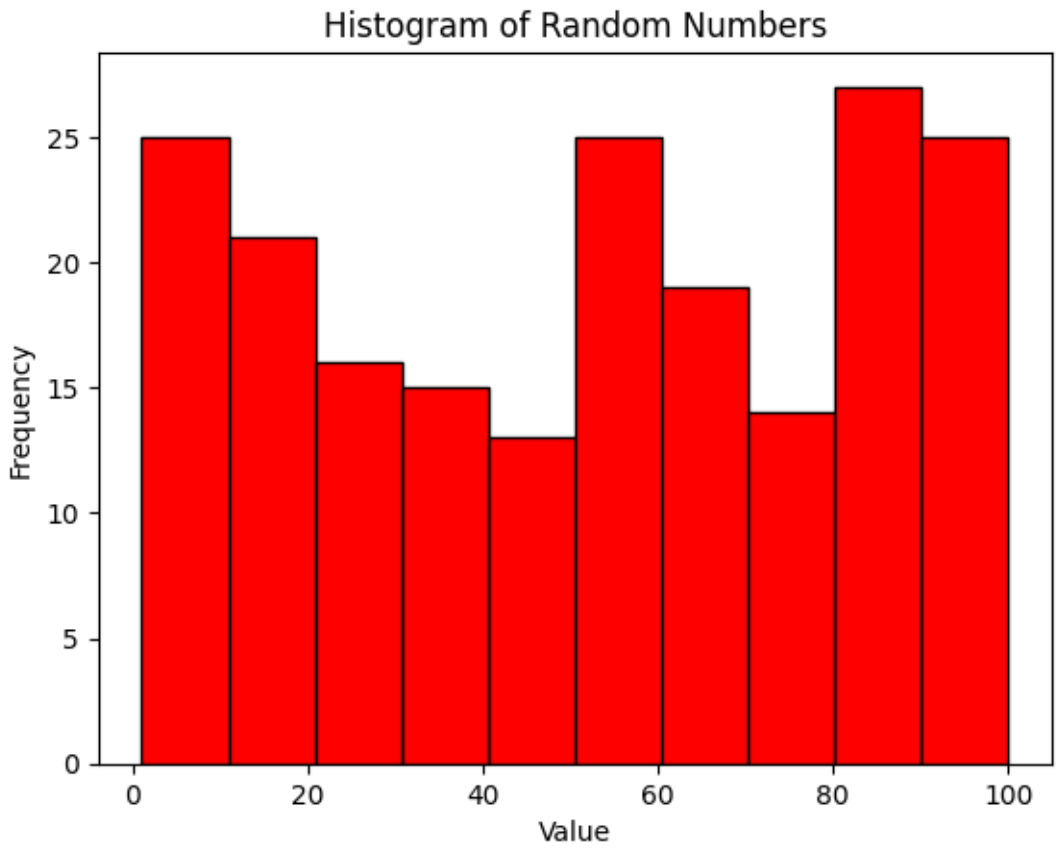
x = list(range(1, 11))
squares = [i ** 2 for i in x]

plt.plot(x, x, marker='o', label='Numbers')
plt.plot(x, squares, marker='s', label='Squares')
plt.title("Numbers vs Their Squares")
plt.xlabel("x")
plt.ylabel("Value")
plt.grid(True)
plt.show()
```



[25]:
```
#question 30
import matplotlib.pyplot as plt
import random

data = [random.randint(1, 100) for _ in range(200)]
plt.hist(data, bins=10, color='red', edgecolor='black')
plt.title("Histogram of Random Numbers")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```


Histogram of Random Numbers
Frequency
Value
0
5
10
15
20
25
0
20
40
60
80
100

[ ]: